

```

import pandas as pd
import numpy as np
from sklearn import linear_model
from sklearn.model_selection import train_test_split
from sklearn import metrics
data= pd.read_csv("iris.csv")
data.head(10)
data.shape

(150, 5)

x=data.drop('Name',axis=1)
y=data['Name']
from sklearn.model_selection import train_test_split
x_train,x_test, y_train,y_test=train_test_split(x,y,test_size=0.20)
logr=linear_model.LogisticRegression()
logr.fit(x_train,y_train)

predicted=logr.predict([[5.0,3.6,1.4,0.2]])

print(predicted)
y_pred=logr.predict(x_test)
print(x_test,y_pred)

```

```

['Iris-setosa']

```

	SepalLength	SepalWidth	PetalLength	PetalWidth
24	4.8	3.4	1.9	0.2
105	7.6	3.0	6.6	2.1
38	4.4	3.0	1.3	0.2
6	4.6	3.4	1.4	0.3
52	6.9	3.1	4.9	1.5
96	5.7	2.9	4.2	1.3
137	6.4	3.1	5.5	1.8
124	6.7	3.3	5.7	2.1
35	5.0	3.2	1.2	0.2
25	5.0	3.0	1.6	0.2
27	5.2	3.5	1.5	0.2
72	6.3	2.5	4.9	1.5
10	5.4	3.7	1.5	0.2
12	4.8	3.0	1.4	0.1
11	4.8	3.4	1.6	0.2
84	5.4	3.0	4.5	1.5
143	6.8	3.2	5.9	2.3
20	5.4	3.4	1.7	0.2
33	5.5	4.2	1.4	0.2
128	6.4	2.8	5.6	2.1
119	6.0	2.2	5.0	1.5
145	6.7	3.0	5.2	2.3
89	5.5	2.5	4.0	1.3
104	6.5	3.0	5.8	2.2

```

60          5.0          2.0          3.5          1.0
78          6.0          2.9          4.5          1.5
18          5.7          3.8          1.7          0.3
115         6.4          3.2          5.3          2.3
69          5.6          2.5          3.9          1.1
53          5.5          2.3          4.0          1.3 ['Iris-setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor' 'Iris-
setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-setosa' 'Iris-virginica' 'Iris-virginica'
'Iris-virginica' 'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor']

```

C:\Users\MGM\anaconda3\Lib\site-packages\sklearn\base.py:493:

UserWarning: X does not have valid feature names, but
LogisticRegression was fitted with feature names
warnings.warn(

```
from sklearn.metrics import classification_report, confusion_matrix
```

```
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
[[12  0  0]
 [ 0  9  0]
 [ 0  0  9]]
```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	12
Iris-versicolor	1.00	1.00	1.00	9
Iris-virginica	1.00	1.00	1.00	9
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```

x=data.drop('Name',axis=1)
y=data['Name']
from sklearn.model_selection import train_test_split
x_train,x_test, y_train,y_test=train_test_split(x,y,test_size=0.30)
logr=linear_model.LogisticRegression()
logr.fit(x_train,y_train)

predicted=logr.predict([[5.0,3.6,1.4,0.2]])
print(predicted)

```

```
y_pred=logr.predict(x_test)
print(x_test,y_pred)
```

```
['Iris-setosa']
SepalLength SepalWidth PetalLength PetalWidth
106          4.9         2.5         4.5         1.7
63           6.1         2.9         4.7         1.4
3            4.6         3.1         1.5         0.2
89           5.5         2.5         4.0         1.3
8            4.4         2.9         1.4         0.2
37           4.9         3.1         1.5         0.1
48           5.3         3.7         1.5         0.2
104          6.5         3.0         5.8         2.2
148          6.2         3.4         5.4         2.3
149          5.9         3.0         5.1         1.8
138          6.0         3.0         4.8         1.8
83           6.0         2.7         5.1         1.6
125          7.2         3.2         6.0         1.8
136          6.3         3.4         5.6         2.4
135          7.7         3.0         6.1         2.3
28           5.2         3.4         1.4         0.2
53           5.5         2.3         4.0         1.3
130          7.4         2.8         6.1         1.9
76           6.8         2.8         4.8         1.4
71           6.1         2.8         4.0         1.3
140          6.7         3.1         5.6         2.4
123          6.3         2.7         4.9         1.8
108          6.7         2.5         5.8         1.8
13           4.3         3.0         1.1         0.1
65           6.7         3.1         4.4         1.4
81           5.5         2.4         3.7         1.0
129          7.2         3.0         5.8         1.6
132          6.4         2.8         5.6         2.2
98           5.1         2.5         3.0         1.1
22           4.6         3.6         1.0         0.2
92           5.8         2.6         4.0         1.2
79           5.7         2.6         3.5         1.0
101          5.8         2.7         5.1         1.9
21           5.1         3.7         1.5         0.4
1           4.9         3.0         1.4         0.2
147          6.5         3.0         5.2         2.0
50           7.0         3.2         4.7         1.4
38           4.4         3.0         1.3         0.2
120          6.9         3.2         5.7         2.3
113          5.7         2.5         5.0         2.0
0            5.1         3.5         1.4         0.2
85           6.0         3.4         4.5         1.6
70           5.9         3.2         4.8         1.8
96           5.7         2.9         4.2         1.3
46           5.1         3.8         1.6         0.2 ['Iris-
```

```

versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor'
'Iris-setosa' 'Iris-setosa' 'Iris-setosa' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-setosa'
'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor' 'Iris-
versicolor'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-virginica'
'Iris-versicolor' 'Iris-setosa' 'Iris-versicolor' 'Iris-versicolor'
'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-virginica'
'Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor'
'Iris-setosa']

```

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```

from sklearn.metrics import classification_report, confusion_matrix

```

```

print(confusion_matrix(y_test,y_pred))

```

```

print(classification_report(y_test,y_pred))

```

```

[[12  0  0]
 [ 0 13  2]
 [ 0  1 17]]

```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	12
Iris-versicolor	0.93	0.87	0.90	15
Iris-virginica	0.89	0.94	0.92	18
accuracy			0.93	45
macro avg	0.94	0.94	0.94	45
weighted avg	0.93	0.93	0.93	45

```

x=data.drop('Name',axis=1)

```

```

y=data['Name']

```

```

from sklearn.model_selection import train_test_split

```

```

x_train,x_test, y_train,y_test=train_test_split(x,y,test_size=0.35)

```

```

logr=linear_model.LogisticRegression()

```

```

logr.fit(x_train,y_train)

```

```

predicted=logr.predict([[5.0,3.6,1.4,0.2]])

```

```

print(predicted)

```

```

y_pred=logr.predict(x_test)

```

```

print(x_test,y_pred)

```

['Iris-setosa']				
	SepalLength	SepalWidth	PetalLength	PetalWidth
87	6.3	2.3	4.4	1.3
56	6.3	3.3	4.7	1.6
65	6.7	3.1	4.4	1.4
142	5.8	2.7	5.1	1.9
31	5.4	3.4	1.5	0.4
85	6.0	3.4	4.5	1.6
78	6.0	2.9	4.5	1.5
133	6.3	2.8	5.1	1.5
97	6.2	2.9	4.3	1.3
134	6.1	2.6	5.6	1.4
74	6.4	2.9	4.3	1.3
9	4.9	3.1	1.5	0.1
23	5.1	3.3	1.7	0.5
12	4.8	3.0	1.4	0.1
70	5.9	3.2	4.8	1.8
5	5.4	3.9	1.7	0.4
54	6.5	2.8	4.6	1.5
76	6.8	2.8	4.8	1.4
108	6.7	2.5	5.8	1.8
37	4.9	3.1	1.5	0.1
130	7.4	2.8	6.1	1.9
140	6.7	3.1	5.6	2.4
83	6.0	2.7	5.1	1.6
2	4.7	3.2	1.3	0.2
84	5.4	3.0	4.5	1.5
100	6.3	3.3	6.0	2.5
27	5.2	3.5	1.5	0.2
79	5.7	2.6	3.5	1.0
43	5.0	3.5	1.6	0.6
77	6.7	3.0	5.0	1.7
117	7.7	3.8	6.7	2.2
147	6.5	3.0	5.2	2.0
53	5.5	2.3	4.0	1.3
1	4.9	3.0	1.4	0.2
36	5.5	3.5	1.3	0.2
88	5.6	3.0	4.1	1.3
61	5.9	3.0	4.2	1.5
66	5.6	3.0	4.5	1.5
15	5.7	4.4	1.5	0.4
18	5.7	3.8	1.7	0.3
123	6.3	2.7	4.9	1.8
17	5.1	3.5	1.4	0.3
104	6.5	3.0	5.8	2.2
50	7.0	3.2	4.7	1.4
98	5.1	2.5	3.0	1.1
26	5.0	3.4	1.6	0.4
22	4.6	3.6	1.0	0.2
58	6.6	2.9	4.6	1.3

```

149         5.9         3.0         5.1         1.8
101         5.8         2.7         5.1         1.9
42          4.4         3.2         1.3         0.2
49          5.0         3.3         1.4         0.2
46          5.1         3.8         1.6         0.2 ['Iris-
versicolor' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-setosa' 'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica'
'Iris-versicolor' 'Iris-virginica' 'Iris-versicolor' 'Iris-setosa'
'Iris-setosa' 'Iris-setosa' 'Iris-virginica' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-setosa'
'Iris-versicolor' 'Iris-virginica' 'Iris-setosa' 'Iris-versicolor'
'Iris-setosa' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-versicolor' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-virginica' 'Iris-versicolor'
'Iris-versicolor' 'Iris-setosa' 'Iris-setosa' 'Iris-versicolor'
'Iris-virginica' 'Iris-virginica' 'Iris-setosa' 'Iris-setosa'
'Iris-setosa']

```

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warnings.warn(

```
from sklearn.metrics import classification_report, confusion_matrix
```

```
print(confusion_matrix(y_test, y_pred))
```

```
print(classification_report(y_test, y_pred))
```

```

[[19  0  0]
 [ 0 18  3]
 [ 0  0 13]]

```

	precision	recall	f1-score	support
Iris-setosa	1.00	1.00	1.00	19
Iris-versicolor	1.00	0.86	0.92	21
Iris-virginica	0.81	1.00	0.90	13
accuracy			0.94	53
macro avg	0.94	0.95	0.94	53
weighted avg	0.95	0.94	0.94	53

```
x=data.drop('Name',axis=1)
```

```
y=data['Name']
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test, y_train,y_test=train_test_split(x,y,test_size=0.40)
```

```
logr=linear_model.LogisticRegression()
```

```
logr.fit(x_train,y_train)
```

```
predicted=logr.predict([[5.0,3.6,1.4,0.2]])
```

```
print(predicted)
```

```
y_pred=logr.predict(x_test)
```

```
print(x_test,y_pred)
```

```
['Iris-setosa']
```

	SepalLength	SepalWidth	PetalLength	PetalWidth
53	5.5	2.3	4.0	1.3
28	5.2	3.4	1.4	0.2
117	7.7	3.8	6.7	2.2
24	4.8	3.4	1.9	0.2
4	5.0	3.6	1.4	0.2
111	6.4	2.7	5.3	1.9
8	4.4	2.9	1.4	0.2
107	7.3	2.9	6.3	1.8
64	5.6	2.9	3.6	1.3
57	4.9	2.4	3.3	1.0
109	7.2	3.6	6.1	2.5
42	4.4	3.2	1.3	0.2
108	6.7	2.5	5.8	1.8
104	6.5	3.0	5.8	2.2
120	6.9	3.2	5.7	2.3
140	6.7	3.1	5.6	2.4
96	5.7	2.9	4.2	1.3
97	6.2	2.9	4.3	1.3
6	4.6	3.4	1.4	0.3
129	7.2	3.0	5.8	1.6
38	4.4	3.0	1.3	0.2
123	6.3	2.7	4.9	1.8
14	5.8	4.0	1.2	0.2
49	5.0	3.3	1.4	0.2
47	4.6	3.2	1.4	0.2
70	5.9	3.2	4.8	1.8
26	5.0	3.4	1.6	0.4
39	5.1	3.4	1.5	0.2
46	5.1	3.8	1.6	0.2
138	6.0	3.0	4.8	1.8
11	4.8	3.4	1.6	0.2
103	6.3	2.9	5.6	1.8
148	6.2	3.4	5.4	2.3
76	6.8	2.8	4.8	1.4
136	6.3	3.4	5.6	2.4
149	5.9	3.0	5.1	1.8
41	4.5	2.3	1.3	0.3
74	6.4	2.9	4.3	1.3
50	7.0	3.2	4.7	1.4
112	6.8	3.0	5.5	2.1
134	6.1	2.6	5.6	1.4
58	6.6	2.9	4.6	1.3

69	5.6	2.5	3.9	1.1
83	6.0	2.7	5.1	1.6
144	6.7	3.3	5.7	2.5
36	5.5	3.5	1.3	0.2
73	6.1	2.8	4.7	1.2
85	6.0	3.4	4.5	1.6
2	4.7	3.2	1.3	0.2
78	6.0	2.9	4.5	1.5
60	5.0	2.0	3.5	1.0
101	5.8	2.7	5.1	1.9
61	5.9	3.0	4.2	1.5
77	6.7	3.0	5.0	1.7
115	6.4	3.2	5.3	2.3
68	6.2	2.2	4.5	1.5
23	5.1	3.3	1.7	0.5
17	5.1	3.5	1.4	0.3
98	5.1	2.5	3.0	1.1
18	5.7	3.8	1.7	0.3

```

'Iris-versicolor' 'Iris-setosa' 'Iris-virginica' 'Iris-setosa'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-setosa'
'Iris-virginica' 'Iris-virginica' 'Iris-virginica' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa' 'Iris-virginica'
'Iris-setosa' 'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-
setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-setosa' 'Iris-setosa'
'Iris-virginica' 'Iris-setosa' 'Iris-virginica' 'Iris-virginica'
'Iris-versicolor' 'Iris-virginica' 'Iris-virginica' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-virginica'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-virginica'
'Iris-setosa' 'Iris-versicolor' 'Iris-versicolor' 'Iris-setosa'
'Iris-versicolor' 'Iris-versicolor' 'Iris-virginica' 'Iris-
versicolor'
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'Iris-setosa' 'Iris-versicolor' 'Iris-setosa']

```

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```
from sklearn.metrics import classification_report, confusion_matrix
```

```
print(confusion_matrix(y_test, y_pred))
```

```
print(classification_report(y_test, y_pred))
```

```
[[20  0  0]
 [ 0 17  3]
 [ 0  0 20]]
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

Iris-setosa	1.00	1.00	1.00	20
Iris-versicolor	1.00	0.85	0.92	20
Iris-virginica	0.87	1.00	0.93	20
accuracy			0.95	60
macro avg	0.96	0.95	0.95	60
weighted avg	0.96	0.95	0.95	60